

```
import pandas as pd
```

```
Q4_promotions = pd.read_excel("/content/Promotion.xlsx")
```

```
Q4_promotions
```

	Merchant Name	GMV EGP	Count of transaction_id	Profit	Prof Marg	GMV_Per	Trans_perc	Profit_Perc	Profit Margin Score	Composite Metric
0	Royal Boutique 715	4593.80	4	58.95	0.012833	0.060	0.034	0.493	1.000000	0.278700
1	Blue Gold 906	5229042.40	798	55549.09	0.010623	0.997	0.960	0.998	0.894425	0.979642
2	Top Boutique 61	447738.26	562	4332.34	0.009676	0.852	0.910	0.949	0.849165	0.892416
3	Gold Interiors 885	8249.50	5	73.92	0.008961	0.082	0.046	0.509	0.814973	0.276197
4	Gold Wear 449	475618.36	624	4254.06	0.008944	0.857	0.917	0.949	0.814196	0.892320
...	...	...	...	...	...	...	...	...	...	...
962	Gold Mart 739	16273.97	40	-89.41	-0.005494	0.400	0.400	0.400	0.124244	0.372424
963	Nile Bites 129	1592.11	6	-8.88	-0.005578	0.000	0.250	0.750	0.120256	0.287026
964	Nile Diner 845	16813.52	56	-101.45	-0.006034	0.333	0.333	0.333	0.098450	0.309545
965	El Mart 154	30342.30	69	-225.51	-0.007432	0.500	0.500	0.000	0.031627	0.303163
966	Al Drive 732	3989.35	1	-32.29	-0.008094	0.000	0.000	0.000	0.000000	0.000000

```
967 rows x 10 columns
```

```
print(Q4_promotions.head())
```

	Merchant Name	GMV EGP	Count of transaction_id	Profit	Prof Marg	GMV_Per	Trans_perc	Profit_Perc	Profit Margin Score	Composite Metric
0	Royal Boutique 715	4593.80	4	58.95	0.012833	0.060	0.034	0.493	1.000000	0.278700
1	Blue Gold 906	5229042.40	798	55549.09	0.010623	0.997	0.960	0.998	0.894425	0.979642
2	Top Boutique 61	447738.26	562	4332.34	0.009676	0.852	0.910	0.949	0.849165	0.892416
3	Gold Interiors 885	8249.50	5	73.92	0.008961	0.082	0.046	0.509	0.814973	0.276197
4	Gold Wear 449	475618.36	624	4254.06	0.008944	0.857	0.917	0.949	0.814196	0.892320

```
print(Q4_promotions.columns)
```

```
Index(['Merchant Name', 'GMV EGP', 'Count of transaction_id', 'Profit',  
      'Prof Marg', 'GMV_Per', 'Trans_perc', 'Profit_Perc',  
      'Profit Margin Score', 'Composite Metric'],  
      dtype='object')
```

```
print(Q4_promotions.dtypes)
```

```
Merchant Name      object  
GMV EGP            float64  
Count of transaction_id  int64  
Profit             float64  
Prof Marg          float64  
GMV_Per            float64  
Trans_perc         float64  
Profit_Perc        float64  
Profit Margin Score float64  
Composite Metric   float64  
dtype: object
```

```
print(Q4_promotions.isnull().sum())
```

```
Merchant Name      0  
GMV EGP            0  
Count of transaction_id  0  
Profit             0  
Prof Marg          0  
GMV_Per            0  
Trans_perc         0  
Profit_Perc        0  
Profit Margin Score 0  
Composite Metric   0  
dtype: int64
```

```
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```

```
Q4_promotions["rank"] = Q4_promotions["Composite Metric"].rank(method="first", ascending=False)
```

```
Q4_promotions
```

	Merchant Name	GMV_EGP	Count of transaction_id	Profit	Prof_Marg	GMV_Per	Trans_perc	Profit_Perc	Profit Margin Score	Composite Metric	rank
0	Royal Boutique 715	4593.80	4	58.95	0.012833	0.060	0.034	0.493	1.000000	0.278700	938.0
1	Blue Gold 906	5229042.40	798	55549.09	0.010623	0.997	0.960	0.998	0.894425	0.979642	1.0
2	Top Boutique 61	447738.26	562	4332.34	0.009676	0.852	0.910	0.949	0.849165	0.892416	110.0
3	Gold Interiors 885	8249.50	5	73.92	0.008961	0.082	0.046	0.509	0.814973	0.276197	940.0
4	Gold Wear 449	475618.36	624	4254.06	0.008944	0.857	0.917	0.949	0.814196	0.892320	111.0
...	...	...	...	...	...	...	...	...	...	...	...
962	Gold Mart 739	16273.97	40	-89.41	-0.005494	0.400	0.400	0.400	0.124244	0.372424	897.0
963	Nile Bites 129	1592.11	6	-8.88	-0.005578	0.000	0.250	0.750	0.120256	0.287026	936.0
964	Nile Diner 845	16813.52	56	-101.45	-0.006034	0.333	0.333	0.333	0.098450	0.309545	930.0
965	El Mart 154	30342.30	69	-225.51	-0.007432	0.500	0.500	0.000	0.031627	0.303163	932.0
966	Al Drive 732	3989.35	1	-32.29	-0.008094	0.000	0.000	0.000	0.000000	0.000000	967.0

967 rows × 11 columns

Q4_promotions.head(26)

	Merchant Name	GMV_EGP	Count of transaction_id	Profit	Prof_Marg	GMV_Per	Trans_perc	Profit_Perc	Profit Margin Score	Composite Metric	rank
0	Royal Boutique 715	4593.80	4	58.95	0.012833	0.060	0.034	0.493	1.000000	0.278700	938.0
1	Blue Gold 906	5229042.40	798	55549.09	0.010623	0.997	0.960	0.998	0.894425	0.979642	1.0
2	Top Boutique 61	447738.26	562	4332.34	0.009676	0.852	0.910	0.949	0.849165	0.892416	110.0
3	Gold Interiors 885	8249.50	5	73.92	0.008961	0.082	0.046	0.509	0.814973	0.276197	940.0
4	Gold Wear 449	475618.36	624	4254.06	0.008944	0.857	0.917	0.949	0.814196	0.892320	111.0
5	Blue Boutique 361	663515.99	841	5867.66	0.008843	0.905	0.972	0.964	0.809370	0.926537	39.0
6	Pro Style 47	1022244.21	1238	8691.32	0.008502	0.952	0.998	0.985	0.793071	0.955207	4.0
7	Pro Wear 251	701353.26	920	5932.85	0.008459	0.917	0.991	0.966	0.791014	0.933901	33.0
8	Mega Threads 740	790031.37	1017	6621.62	0.008381	0.935	0.994	0.973	0.787301	0.943430	20.0
9	Royal Wear 24	818116.43	1089	6855.35	0.008379	0.939	0.998	0.975	0.787204	0.946420	14.0
10	Star Boutique 995	222581.80	292	1860.42	0.008358	0.754	0.882	0.902	0.786198	0.827220	204.0
11	Top Style 522	237832.38	308	1972.97	0.008296	0.768	0.893	0.908	0.783200	0.836520	192.0
12	Smart Home 422	770459.44	701	6358.20	0.008252	0.935	0.939	0.972	0.781138	0.931514	36.0
13	City Boutique 762	511637.03	663	4142.20	0.008096	0.874	0.936	0.954	0.773659	0.900366	93.0
14	Prime Wear 517	684604.18	884	5441.44	0.007948	0.917	0.988	0.968	0.766602	0.931460	37.0
15	Smart Casa 332	954688.67	830	7525.54	0.007883	0.946	0.976	0.980	0.763468	0.943947	15.0
16	Gold Threads 818	838849.74	1080	6481.65	0.007727	0.941	0.998	0.974	0.756019	0.943802	16.0
17	Al Fit 895	221179.57	224	1707.81	0.007721	0.756	0.788	0.900	0.755758	0.805576	238.0
18	Al Explore 12	3015183.69	639	22864.50	0.007583	0.992	0.932	0.996	0.749152	0.956915	3.0
19	Prime Style 417	153879.37	204	1164.33	0.007567	0.646	0.757	0.857	0.748358	0.741736	329.0
20	Smart Gems 797	4612597.00	744	34826.80	0.007550	0.995	0.955	0.998	0.747587	0.963159	2.0
21	Royal Boutique 393	4948.94	7	37.29	0.007535	0.064	0.066	0.463	0.746850	0.252385	948.0
22	Prime Sports 362	114894.27	139	864.42	0.007524	0.588	0.633	0.813	0.746308	0.680331	412.0
23	El Living 420	305387.46	290	2214.73	0.007252	0.817	0.887	0.926	0.733338	0.855334	170.0
24	Top Fashion 289	3276.44	6	23.75	0.007249	0.048	0.056	0.426	0.733172	0.231517	956.0
25	Al Boutique 144	265610.36	331	1897.32	0.007143	0.796	0.909	0.914	0.728132	0.847213	182.0

```
import seaborn as sns
import matplotlib.pyplot as plt

cols = ["GMV_EGP", "Profit", "Prof_Marg", "Count of transaction_id"]

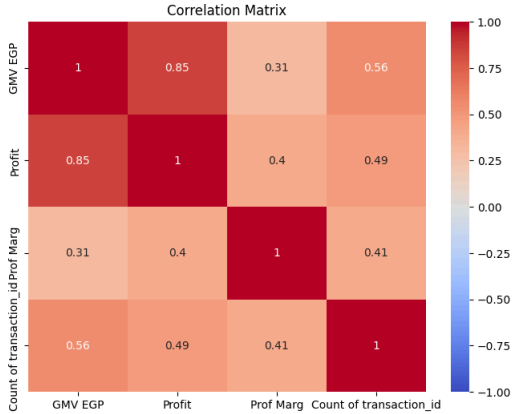
corr = Q4_promotions[cols].corr()

print(corr)

plt.figure(figsize=(8,6))
sns.heatmap(corr, annot=True, cmap="coolwarm", vmin=-1, vmax=1)
plt.title("Correlation Matrix")
plt.show()
```

GMV EGP GMV EGP Profit Prof Marg \
GMV EGP 1.000000 0.850614 0.305203 \
Profit 0.850614 1.000000 0.402388 \
Prof Marg 0.305203 0.402388 1.000000 \
Count of transaction_id 0.556796 0.488850 0.408385

Count of transaction_id
GMV EGP 0.556796
Profit 0.488850
Prof Marg 0.408385
Count of transaction_id 1.000000



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```
Q4_promotions = Q4_promotions.sort_values(by="Composite Metric", ascending=False)
```

Q4_promotions.head(20)

	Merchant Name	GMV_EGP	Count of transaction_id	Profit	Prof_Marg	GMV_Per	Trans_perc	Profit_Perc	Profit Margin Score	Composite Metric	rank	
1	Blue Gold 906	5229042.40		798	55549.09	0.010623	0.997	0.960	0.998	0.894425	0.979642	1.0
20	Smart Gems 797	4612597.00		744	34826.80	0.007550	0.995	0.955	0.998	0.747587	0.963159	2.0
18	Al Explore 12	3015183.69		639	22864.50	0.007583	0.992	0.932	0.996	0.749152	0.956915	3.0
6	Pro Style 47	1022244.21		1238	8691.32	0.008502	0.952	0.998	0.985	0.793071	0.955207	4.0
118	Delta Luxe 183	5146843.71		788	24904.31	0.004839	0.997	0.975	0.998	0.618009	0.955001	5.0
151	Pro Drive 555	2874414.22		864	12864.78	0.004476	0.992	0.990	0.996	0.600656	0.953666	6.0
142	Top World 298	2332059.54		896	10671.16	0.004576	0.991	0.993	0.993	0.605446	0.953445	7.0
38	Delta World 50	1942167.73		730	12729.99	0.006555	0.982	0.953	0.994	0.699999	0.951600	8.0
152	Top Travel 242	3304959.74		754	14748.66	0.004463	0.996	0.970	0.997	0.600033	0.951503	9.0
256	Smart Store 330	2892845.11		1074	9467.14	0.003273	0.995	0.998	0.997	0.543169	0.951017	10.0
240	City Gold 402	5291855.60		813	17925.13	0.003387	0.998	0.984	0.998	0.548650	0.950265	11.0
359	Top Store 132	2331856.71		905	5358.53	0.002298	0.998	0.998	0.998	0.496595	0.947859	12.0
158	City Tech 602	2095352.85		799	9056.83	0.004322	0.986	0.978	0.995	0.593332	0.947833	13.0
9	Royal Wear 24	818116.43		1089	6855.35	0.008379	0.939	0.998	0.975	0.787204	0.946420	14.0
15	Smart Casa 332	954688.67		830	7525.54	0.007883	0.946	0.976	0.980	0.763468	0.943947	15.0
16	Gold Threads 818	838849.74		1080	6481.65	0.007727	0.941	0.998	0.974	0.756019	0.943802	16.0
339	Top Tech 709	2169675.82		807	5228.28	0.002410	0.995	0.985	0.995	0.501934	0.943693	17.0
72	Delta Living 720	1052479.11		986	6014.20	0.005714	0.960	0.996	0.981	0.659849	0.943485	18.0
27	Nile Casa 559	960240.05		874	6747.24	0.007027	0.950	0.989	0.978	0.722559	0.943456	19.0
8	Mega Threads 740	790031.37		1017	6621.62	0.008381	0.935	0.994	0.973	0.787301	0.943430	20.0

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Percentiles Measure

```
Q4_promotions["Composite Metric"].quantile([0.025,0.5, 0.75, 0.9])
```

Composite Metric	
0.025	0.267539
0.500	0.638261
0.750	0.799944
0.900	0.896719

dtype: float64

p50 = 0.638261
p75 = 0.799944
p90 = 0.896719

```
def assign_tier(score):  
    if score >= p90:  
        return "Platinum"  
    elif score >= p75:  
        return "Gold"  
    elif score >= p50:  
        return "Silver"  
    else:  
        return "Bronze"
```

Q4_promotions["tier"] = Q4_promotions["Composite Metric"].apply(assign_tier)

Q4_promotions["tier"].value_counts()

count	
tier	
Bronze	484
Silver	241
Gold	145
Platinum	97

dtype: int64

Q4_promotions

	Merchant Name	GMV EGP	Count of transaction_id	Profit	Prof Marg	GMV_Per	Trans_perc	Profit_Perc	Profit Margin Score	Composite Metric	rank	tier
1	Blue Gold 906	5229042.40	798	55549.09	0.010623	0.997	0.960	0.998	0.894425	0.979642	1.0	Platinum
20	Smart Gems 797	4612597.00	744	34826.80	0.007550	0.995	0.955	0.998	0.747587	0.963159	2.0	Platinum
18	AI Explore 12	3015183.69	639	22864.50	0.007583	0.992	0.932	0.996	0.749152	0.956915	3.0	Platinum
6	Pro Style 47	1022244.21	1238	8691.32	0.008502	0.952	0.998	0.985	0.793071	0.955207	4.0	Platinum
118	Delta Luxe 183	5146843.71	788	24904.31	0.004839	0.997	0.975	0.998	0.618009	0.955001	5.0	Platinum
...	...	...	...	...	...	...	...	...	...	...	...	...
215	Top Home 386	2251.57	2	8.07	0.003584	0.039	0.007	0.469	0.558057	0.213506	963.0	Bronze
227	Smart Grill 776	1779.77	3	6.23	0.003500	0.033	0.024	0.467	0.554057	0.213506	964.0	Bronze
333	Delta Sports 710	749.89	1	1.86	0.002480	0.011	0.000	0.512	0.505311	0.208531	965.0	Bronze
200	Mega Wellness 545	912.76	1	3.42	0.003747	0.011	0.000	0.438	0.565832	0.192383	966.0	Bronze
966	AI Drive 732	3989.35	1	-32.29	-0.008094	0.000	0.000	0.000	0.000000	0.000000	967.0	Bronze

967 rows × 12 columns

result = Q4_promotions[["rank", "Merchant Name", "GMV EGP", "tier"]]

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result

rank

Merchant Name

GMV EGP

tier

result = result.reset_index(drop=True)

result

rank	Merchant Name	GMV EGP	tier
0	Blue Gold 906	5229042.40	Platinum
1	Smart Gems 797	4612597.00	Platinum
2	AI Explore 12	3015183.69	Platinum
3	Pro Style 47	1022244.21	Platinum
4	Delta Luxe 183	5146843.71	Platinum
...	...	...	...
962	Top Home 386	2251.57	Bronze
963	Smart Grill 776	1779.77	Bronze
964	Delta Sports 710	749.89	Bronze
965	Mega Wellness 545	912.76	Bronze
966	AI Drive 732	3989.35	Bronze

967 rows x 4 columns

result["rank"] = result["rank"].astype(int)

result

rank	Merchant Name	GMV EGP	tier
0	Blue Gold 906	5229042.40	Platinum
1	Smart Gems 797	4612597.00	Platinum
2	AI Explore 12	3015183.69	Platinum
3	Pro Style 47	1022244.21	Platinum
4	Delta Luxe 183	5146843.71	Platinum
...	...	...	...
962	Top Home 386	2251.57	Bronze
963	Smart Grill 776	1779.77	Bronze
964	Delta Sports 710	749.89	Bronze
965	Mega Wellness 545	912.76	Bronze
966	AI Drive 732	3989.35	Bronze

967 rows x 4 columns

result.to_excel("merchant_tiers.xlsx", index=False)

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